INCH-POUND

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MILITARY SPECIFICATION

# DELAY ELEMENT, CARTRIDGE, M69 ASSEMBLY

*This specification is approved for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers requirements for the M69 Cartridge Delay Element (see 6.1).

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

**2.1.1 Specifications, standards, and handbooks.** The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-B-550 | Barium Chromate |
| MIL-A-2550 | Ammunition, General Specification for |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloy |
| MIL-B-51092 | Baron, Amorphous, Powder |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | Quality Assurance Terms and Definitions |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 3730), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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MIL-STD-453 Inspection, Radiographic

MIL-STD-810 Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

**2.1.2 Other Government documents, drawings, and publications.** The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL ORDNANCE STATION (CAGE Codes 19200 and 19203)

DL10551589 Delay Element, Cartridge, M69, Assembly

8826729 Box, Packing, Ammunition, for Delay Elements

8864162 Container, Metal, for Delay Elements

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5035.)

**2.2 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified in the contract or purchase order, a first article sample shall be subjected to first article inspection (see 4.4, 6.2, and 6.3).

**3.2 Conformance to documents.** The M69 delay element covered by this specification shall be manufactured in accordance with DL10551589 and all documents listed thereon.

**3.3 Materials.** All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.



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**3.4 Components.**

**3.4.1 Primary components.** For the purpose of this specification, the M42 Primer, Delay Charge and the output material charge are considered primary components. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot (see 6.7).

**3.4.1.1 Primer.** The primer shall conform to the applicable drawing and reference specification. Only percussion primers from lots manufactured within the last 24 months or recertified within the last 12 months shall be used in these units. Prior to assembly into a delay element, the primer shall be visually examined in accordance with 4.6.5 and radiographically examined for defects as specified in 4.7.1.

**3.4.2 Protective finish.** The finish of each component part shall be in accordance with MIL-A-8625 and the applicable drawing (see 4.7.2).

**3.5 Assembly.**

**3.5.1 Radiographic examination.** Each delay element assembly shall be radiographically examined for defects (see 4.7.3).

**3.5.2 Vibration.** The assembled delay element shall withstand vibration as prescribed in MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I-3.2.5 without evidence of any disintegration of the primer, propellant, or igniter, or derangement of any component (see 4.7.4 and 6.2).

**3.6 Functional requirements.**

**3.6.1 Delay element performance.**

**3.6.1.1 Misfires.** The ignition delay element shall not fail to fire.

**3.6.1.2 Ballistics.** The ignition delay charge shall consist of barium chromate, as specified in MIL-B-550, and boron, as specified in MIL-B-51092, in the proportion and quantity fixed by the contractor to produce the ignition delay times required in table I (see 6.5.1).

TABLE I. Ballistic requirements.

| Temperature (• F) | Ignition delay time (sec) | |
|---|---|---|
| | Lower | Upper |
| -60 to -70 | 1.200 | 1.600 |
| 65 to 75 | 1.300 | 1.500 |
| 195 to 205 | 1.100 | 1.500 |



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**3.7 Workmanship.** Delay elements shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

**3.7.1 Metal defects.** All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

**3.7.2 Burr.** No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

**3.7.3 Foreign matter.** No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

**4.1.1 Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

**4.1.2 Quality assurance terms and definitions.** Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

**4.1.3 Inspections.** Inspections shall be in accordance with MIL-A-2550 and MIL-STD-453.

**4.2 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

**4.3 Inspection conditions.** Unless otherwise specified in the contract or purchase order (see 6.2),



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inspection conditions shall be as follows:

a. Temperature: Room ambient 20• to 25• C (68• to 77• F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric: 14.7 psia

**4.4 First article inspection.** At the beginning of regular production, the contractor shall submit a first article sample to a Government approved facility for evaluation. The sample shall consist of 10 delay bodies, 10 primer retainers, 21 units for ballistics, to be tested as indicated in table II, and 10 sets of unassembled components (including explosive components) for visual and dimensional examination. The first article sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. When it becomes necessary to make a change in either the primer, propellant, or igniter lot after initial production, 12 assembled delay elements from the first production lot thereafter shall be tested in accordance with 4.7.5. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).

TABLE II. First article inspection.

| Paragraph | | Test | Sample size | Type of test | |
|---|---|---|---|---|---|
| Requirement | Test | | | Destructive | Nondestructive |
| 3.4.2 | 4.7.2 | Protective finish | 10 Delay bodies | X | - |
| | | | 10 Primer retainers | X | - |
| 3.5.2 | 4.7.4 | Vibration | 12 (from ballistics sample) | - | X |
| 3.6 | 4.7.5 | Function tests | 21 | X | - |

**4.4.1 Examination, tests, and facility.** After provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract (see 6.2).

**4.4.2 Temperature conditioning for ballistic test.** The sample shall be temperature conditioned and tested to determine compliance with 3.6.1.2. The sample for firing shall be conditioned to table III.

TABLE III. Temperature conditioning.

| Temperature (• F) | Not vibrated | Vibrated |
|---|---|---|
| -60 to -70 | 3 | 4 |



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| | | |
|---|---|---|
| 65 to 75 | 3 | 4 |
| 195 to 205 | 3 | 4 |

**4.4.3 First article failure.** Failure of the sample to comply with the requirements of the drawings and specifications shall result in first article sample disapproval.

**4.5 Quality conformance inspection.**

**4.5.1 Lot.**

**4.5.1.1 Submission of product.** The product shall be submitted in accordance with MIL-STD-105.

**4.5.1.2 Lot size.** Unless otherwise specified (see 6.2), the production lot size shall be not more than 1200.

**4.5.2 Examination and tests.** The quality conformance sample shall be subjected to the examinations of 4.6 and all the test procedures of 4.7 except for 4.7.4.

**4.6 Examination.** All delay elements and components shall be examined visually and radiographically and characteristics defined in accordance with the applicable drawings. One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter. When compliance is in doubt as a result of visual inspection, the characteristic shall be measured or gaged to determine acceptability. All nonconforming items shall be rejected.

**4.6.1 Delay Element, M69, Assembly Drawing 10551589.**

| Characteristic | Defect | Method of Inspection |
|---|---|---|
| Critical | | |
| 1 | Any part missing | X-ray/Visual |
| 2 | Improper assembly | X-ray/Visual |
| 3 | Primer not properly seated | |



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| | | |
|---|---|---|
| 4 | Mixed ammunition | X-ray/Visual |
| 5 | Flash hole missing or obstructed | Visual |
| 6 | Primer uncrimped or seal missing | Visual |
| 7 | Marking missing, incorrect or illegible | Visual |
| 8 | Delay charge not visible through flash hole | Visual |
| Major | | |
| 101 | Concentricity of flange with body | Gage |
| 102 | Diameter of body, max | Gage |
| 103 | Retainer cup not properly seated | Visual |
| 104 | Firing plug not flush or below flush with retaining cup | Visual |
| 105 | Metal defective | Visual |
| 106 | Protective finish inadequate or defective | Visual |
| 107 | Burr | Visual |
| 108 | Foreign matter | Visual |



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| | | |
|---|---|---|
| | | Visual |
| Minor | | |
| 201 | Total length, max | |
| 202 | Lacquer missing or inadequate | Gage<br>Visual |

**4.6.2 Body Delay, Drawing 8594081, covering a detail of Drawing 10551589.**

| | | |
|---|---|---|
| Critical | | |
| 1 | Flash hole missing or obstructed | |
| 2 | Metal defective | Visual<br>Visual |
| Major | | |
| 101 | Pitch diameter, external thread, min | |
| 102 | Major diameter, external thread, min | Gage |
| 103 | Concentricity of body with pitch diameter of thread | Gage<br>Gage |



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| | | |
|---|---|---|
| 104 | Diameter of cavity, max | Gage |
| 105 | Diameter of head, min | Gage |
| 106 | Diameter of flash hole | Gage |
| 107 | Diameter of thread relief | Gage |
| 108 | Length from cavity face to shoulder of head diameter | Gage |
| 109 | Protective finish inadequate or defective | Visual |
| 110 | Thread damaged or not full | Visual |
| 111 | Burr | Visual |
| 112 | Foreign matter present | |



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| | | |
|---|---|---|
| | Visual | |
| Minor<br>201 | Pitch diameter of external thread, max | |
| 202 | Gage<br>Diameter of wrench hole | |
| 203 | Diameter of thread pilot | Gage |
| 204 | Diameter of cavity, min | Gage |
| 205 | Gage<br>Diameter of head, max | |
| 206 | Gage<br>Diameter of step rear of head | |
| 207 | Gage<br>Depth of wrench hole | |
| 208 | Gage<br>Depth to base of cavity | |
| 209 | Gage<br>Total length | |



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| | | |
|---|---|---|
| 210 | Length of step rear of head | Gage |
| 211 | Length of thread pilot | Gage |
| 212 | Width of thread relief | Gage |
| 213 | Location of wrench hole | Gage |
| 214 | Radius or chamfer missing or incorrect | Gage |
| 215 | Surface finish improper | Visual |

**4.6.3 Gasket, Aluminum, Drawing 8594082, covering a detail of Drawing 10551589.**

Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Diameter, outside, min | Visual |



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| | | |
|---|---|---|
| 102 | Diameter, inside, max | Gage |
| 103 | Metal defective | Gage |
| 104 | Burr | Visual |
| 105 | Foreign matter present | Visual |
| Minor | | Visual |
| 201 | Diameter, outside, max | |
| 202 | Diameter, inside, min | Gage |
| 203 | Thickness | Gage |



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| | | |
|---|---|---|
| 204 | Surface finish improper | Gage |
| | | Visual |

**4.6.4 Primer, Retainer, Drawing 10520003, covering a detail of Drawing 10551589.**

| | | |
|---|---|---|
| Critical | | |
| 1 | Depth of head counterbore | Gage |
| 2 | Depth of primer pocket | Gage |
| 3 | Depth of cup pocket | Gage |
| 4 | Metal defects | Visual |
| Major | | |
| 101 | Pitch diameter of internal thread, max | Gage |
| 102 | Minor diameter of internal thread, max | Gage |
| 103 | Concentricity of flange with primer pocket | Gage |
| 104 | Concentricity of flange hole with primer pocket | Gage |
| 105 | Concentricity of large end with primer pocket | Gage |
| 106 | Concentricity of head counterbore with primer pocket | Gage |
| 107 | Concentricity of cup pocket with primer pocket | Gage |
| 108 | Concentricity of body with primer pocket | Gage |
| 109 | Concentricity of pitch diameter of thread with body | Gage |
| 110 | Symmetry wrench flats with pitch diameter of thread | Gage |
| 111 | Diameter of body, min | Gage |
| 112 | Diameter of primer pocket, min | Gage |
| 113 | Diameter of cup pocket, min | Gage |
| 114 | Diameter of head counterbore | Gage |
| 115 | Depth of primer pocket | Gage |
| 116 | Depth of cup pocket | Gage |
| 117 | Length of effective thread, min | Gage |
| 118 | Location of flange | Gage |
| 119 | Location of datum diameter | Gage |
| 120 | Thread damaged or not full | Visual |
| 121 | Protective finish inadequate or defective | Visual |
| 122 | Surface finish of primer pocket improper | Visual |
| 123 | Burr or sharp corner | Visual |
| 124 | Foreign matter | Visual |
| Minor | | |



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| | | |
|---|---|---|
| 201 | Pitch diameter of internal thread, min | Gage |
| 202 | Diameter of body, max | Gage |
| 203 | Diameter of primer pocket, max | Gage |
| 204 | Diameter of cup pocket, max | Gage |
| 205 | Diameter of head | Gage |
| 206 | Depth of wrench flats | Gage |
| 207 | Depth to bottom of threaded cavity | Gage |
| 208 | Length of wrench flats | Gage |
| 209 | Length across wrench flats | Gage |
| 210 | Total length | Gage |
| 211 | Location of wrench flats | Gage |
| 212 | Radius or chamfer missing or incorrect | Gage |
| 213 | Surface finish improper, except as otherwise classified | Visual |

**4.6.5 Primer Assembly, Percussion, Drawing 5184975, covering a detail of Drawing 10551589.**

| | | |
|---|---|---|
| Critical | | |
| 1 | Mixed primers | Visual |
| 2 | Primer pellet broken, cracked, or dislodged | Visual |
| 3 | Primer foil discolored | Visual |
| 4 | Metal defective | Visual |
| Major | | |
| 101 | Burr | Visual |



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| | | |
|---|---|---|
| | | Visual |
| 102 | Foreign matter present | |
| | | Visual |

Minor
None defined

**4.6.6 Cup, Retaining, Drawing 8597232, covering a detail of Drawing 10551589.**

| | | |
|---|---|---|
| Critical | | |
| 1 | Metal defective | Visual |
| Major | | |
| 101 | Concentricity of large cavity with small cavity | Gage |
| 102 | Concentricity of body with small cavity | Gage |
| 103 | Diameter of large cavity, max | Gage |
| 104 | Diameter of small cavity, max | Gage |
| 105 | Diameter of body, min | Gage |
| 106 | Depth of large cavity | Gage |
| 107 | Surface finish of small cavity and body | Visual |
| 108 | Burr or sharp corner | Visual |
| 109 | Foreign matter | Visual |
| Minor | | |
| 201 | Diameter of large cavity, min | Gage |
| 202 | Diameter of small cavity, min | Gage |
| 203 | Diameter of body, max | Gage |
| 204 | Total length | Gage |
| 205 | Surface finish, except as otherwise classified | Visual |

**4.6.7 Plug, Firing, Drawing 8597234, covering a detail of Drawing 10551589.**

| | | |
|---|---|---|
| Critical | | |
| 1 | Metal defective | Visual |
| Major | | |
| 101 | Concentricity of flange with body | Gage |
| 102 | Diameter of flange, min | Gage |
| 103 | Diameter of body, min | Gage |
| 104 | Length of striker | Gage |
| 105 | Radius of striker | Gage |
| 106 | Surface finish of body improper | Visual |
| 107 | Burr or sharp corner | Visual |



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| | | |
|---|---|---|
| 108 | Foreign matter | Visual |
| Minor | | |
| 201 | Diameter of flange, max | Gage |
| 202 | Diameter of body, max | Gage |
| 203 | Total length | Gage |
| 204 | Location of flange | Gage |
| 205 | Surface finish improper, except as otherwise classified | Visual |

**4.6.8 Container, Metal, for Delay Elements, Drawing 8864162.**

Critical
None defined

Major
None defined

| | | |
|---|---|---|
| Minor | | |
| 201 | Contents loose | Manual/Visual |
| 202 | Separator or filler missing | Visual |
| 203 | Sealing inadequate, incompetent, or incorrect | Visual |
| 204 | Container cut or damaged | Visual |
| 205 | Marking missing, incorrect, or illegible | Visual |

**4.6.9 Box, Packing, Ammunition, for Delay Elements.**

Critical



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None defined

Major
None defined

Minor

| | | |
|---|---|---|
| 201 | Contents loose | Manual/Visual |
| 202 | Hardware missing, loose, broken, or improperly engaged | Visual |
| 203 | Strapping missing, loose, broken, or improperly assembled | Visual |
| 204 | Contents of box exposed | Visual |
| 205 | Box damaged | Visual |
| 206 | Marking missing, incorrect, or illegible | Visual |
| 207 | Number of cartons incorrect | Visual |

**4.7 Test methods and procedures.**

**4.7.1 Primer x-ray.** Prior to assembly into the cartridge, the primer shall be radiographically examined in accordance with MIL-STD-453. Each primer shall be x-rayed on boards or trays with the cup side perpendicular to the rays. The x-ray negatives shall be identified by item nomenclature and by lot number. Primers are to be left mounted to the boards or trays until after the x-rays have been reviewed and defects identified to facilitate removal of defective units. After screening, the x-ray negatives with defectives



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identified shall be forwarded to the engineering activity designated by the procuring activity (see 6.2). Interpretation of the radiographic films shall verify the following:

Presence of Primer Charge
No Material Defects

**4.7.2 Protective finish.** Unless otherwise specified (see 6.2), sampling plans and procedures in the determination of acceptability of coated parts submitted by the contractor shall be in accordance with the contract and MIL-A-8625. The anodized part shall be sectioned longitudinally to better expose the critical surfaces to salt spray test.

**4.7.3 Delay element x-ray.** Prior to assembly into a cartridge, the delay element shall be radiographically examined in accordance with MIL-STD-453. Each delay element shall be x-rayed with the sides of the delay element perpendicular to the rays. The x-ray negatives shall be identified by item nomenclature and by lot number. Delay elements are to be left mounted to the boards or trays until after the x-rays have been reviewed and defects identified to facilitate removal of defective units. After screening, the x-ray negatives with defectives identified shall be forwarded to the engineering activity designated by the procuring activity.

**4.7.4 Vibration test.** Twelve of the 21 assembled delay elements of the first article sample, x-rayed as directed in 4.7.3 and properly identified, shall be subjected to a vibration test. The method and procedures for performing the vibration test shall be in accordance with MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I-3.2.5 (see 6.2). Then the 12 vibrated delay elements shall be x-rayed again. Evidence of any disintegration of the primer, propellant, or derangement of any component shall be cause for rejection of the sample. When there is a change in the method of charging the delay element or in the lot of primers, twelve additional delay elements shall be selected and tested as indicated herein.

**4.7.5 Functional.** The following tests shall be performed by the Government.

**4.7.5.1 Ballistics.**

**4.7.5.1.1 First article.** Twenty one assembled delay elements of the first article sample shall be grouped and temperature conditioned as described in table III and 4.8. Following removal from the conditioning chamber, the delay element shall be tested as required by 3.6.1.2. Failure of delay element to comply with 3.6.1.2 shall be cause for sample rejection.

**4.7.5.1.2 Production lot acceptance.** Twenty one assembled delay elements from each production lot shall be grouped and temperature conditioned in accordance with 3.6.1.2. Failure of a delay element to comply with 3.6.1.2 shall be cause for rejection of the lot. In addition to the sample of delay elements for the above test, 10 additional delay elements shall be furnished from each production lot for investigative purposes.

**4.7.5.2 Retest.** There shall be no retests.

**4.7.5.2.1 Test failure.** If a test failure is attributable to an assignable cause, excluding the test cartridge, the original test results shall be discarded and that test reconducted.

**4.8 Temperature conditioning.** The conditioning chamber shall be preconditioned to the specified temperature. Once the temperature inside the chamber has stabilized, the samples shall be placed inside and maintained at the specified temperature for 4 hours minimum, 24 hours maximum.



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**4.8.1 Reconditioning.** The sample must be tested within five minutes after removal from the conditioning chamber. If this time is exceeded, the cartridge shall be reconditioned in accordance with 4.8.

**4.9 Inspection of packaging.** The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

## 5. PACKAGING

**5.1 Packing and marking.** Packing and marking shall be in accordance with Drawings 8826729 and 8864162, except that the marking shall refer to delay element, cartridge, M69 assembly.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

**6.1 Intended use.** The assembly is used in cartridge actuated devices for Aircrew Emergency Escape Systems.

**6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, date, and revision letter of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced and that MIL-STD-810D shall be used (see 2.1, 3.5.2, and 4.7.4)

c. Whether a first article sample is required (see 3.1, 4.4, and 6.3)

d. Test facility for first article tests (see 4.4.1)

e. Production test activity and lot size (see 4.1 and 4.5.1.2)

f. Any special test conditions (see 4.3)

g. Directions for shipping radiographs of primers and of entire lot of delay element assemblies (see 4.7.1 and 4.7.3)

h. Directions for sampling for protective finish tests (see 4.7.2)

i. Level of preservation and packing (see 5.1)

j. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," (DOD 4145.26M) are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," (OP 5) are applicable.



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**6.3 First article.** When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

**6.4 Hazard notice.** The cartridge described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

**6.5 Definition.**

**6.5.1 Ignition delay time.** Ignition delay time is the time from primer impact to the start of pressure rise.

**6.6 Manufacture as a component.** When the delay element is being manufactured in conjunction with a delay cartridge as a component thereof, the first article requirements for ballistic, vibration, x-ray, and packaging requirements are waived, and the requirements of the cartridge specification shall apply.

**6.7 Subject term (keyword) listing.**

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

**6.8 Changes from previous issue.** Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy-OS

Air Force-99

Review activity:
Air-Force 70

Preparing activity

Navy-OS
(Project 1377-0D77)



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